Paper ID : 61028

BCA (Semester-I) Examination, 2022

(Session : 2021-24)

COMPUTER APPLICATION

[Paper Code : BCA-101]

(Mathematical Foundation)

Time : Three Hours [Maximum Marks : 50]

Note: *Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.*

1. (a) For what value of x, the matrix

$\begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$ is singular?

(b) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, find the value of $A^2 - 4A + 3I$, where I is the unit matrix with proper dimension.

2. (i) If $y = \sin(m \sin^{-1} x)$, prove that
$(1-x^2)y'' + m^2 y = 0$.

(ii) If $y^{\frac{1}{m}} + y^{\frac{1}{m}} = 2x$, prove that
$(x^2-1)y'' + xy' - m^2 y = 0$.

3. If $y = e^{a \sin^{-1} x}$, prove that
$(1-x^2)y'' - xy' - a^2 y = 0$.

4. If $u = \sqrt{x^2+y^2+z^2}$, prove that
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{2}{u}$

5. Evaluate $I = \int \frac{\tan x \cdot \sec^2 x}{1-\tan^2 x} dx$

6. Evaluate double integral $I = \int_0^2 \int_0^2 (x^2+y^2) dx\, dy$

7. Solve the following:
$$(x^2 - y \cdot x^2)\, dy + (y^2 + x \cdot y^2)\, dx = 0$$

8. Apply Maclaurin's theorem to obtain the expansion of $\sec x$.

9. Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 2e^{3x}$

10. Write short notes on the following:
(a) Transpose matrix
(b) Singular matrix
(c) Taylor's theorem
(d) Inverse matrix

------x------